Who Owns a Machine-Verified Proof? Legal and Intellectual Property Questions from the Cathedral Project

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Abstract

The Cathedral—a machine-verified reduction of the Riemann Hypothesis—was built by a human with AI assistance. This paper examines the legal and intellectual property questions raised by this mode of production: (1) copyright and ownership of AI-assisted mathematical proofs, (2) the legal status of AI as co-author or co-inventor, (3) machine-verified proofs as legal evidence, (4) open-source licensing of formal mathematics, and (5) liability for AI-generated mathematical errors. We do not provide legal advice. We identify the questions that existing law does not clearly answer and suggest that these questions will become urgent as AI-assisted formal verification becomes commonplace. This paper is written for attorneys, IP professionals, and policymakers who may encounter these issues in practice.

Contents

1 The Facts of the Case

The Cathedral was produced under the following circumstances:

- **Human contributor**: One individual (the author), a computer scientist, providing architectural design, mathematical strategy, quality control, and editorial judgment.
- **AI contributors**: Two commercial AI systems (Google DeepMind's Gemini and Anthropic's Claude) providing code generation, proof strategy suggestions, and mathematical analysis. Both were used through standard commercial APIs and subscriptions.
- **Verification**: The Lean 4 theorem prover, an open-source software system, verified every logical step. The Mathlib library (open-source, Apache 2.0 license) provided foundational mathematics.
- **Output**: 78 Lean source files, 96 archived files, and 22 companion papers. Approximately 22,670 lines of verified proof code and 60,000 words of documentation.
- **Timeline**: 23 days of development.

2 Question 1: Who Owns the Proof?

2.1 Copyright in Mathematical Proofs

Under U.S. copyright law (17 U.S.C. § 102), copyright protects "original works of authorship fixed in any tangible medium of expression." Mathematical proofs have an ambiguous status:

- The *mathematical content* (theorems, identities, logical relationships) is not copyrightable. Mathematical facts are discoveries, not inventions.
- The *expression* of a proof (the specific sequence of words, the pedagogical choices, the narrative structure) is copyrightable.
- *Computer code* implementing a proof is copyrightable as a literary work.

The Cathedral's Lean code is original expression implementing mathematical content. The code itself is copyrightable; the mathematical results it proves are not.

2.2 AI-Generated Content

The U.S. Copyright Office has consistently held that copyright requires a human author. In *Tháler v. Perlmutter* (2023), the D.C. District Court upheld the Copyright Office's refusal to register a work created autonomously by AI.

However, the Cathedral is not autonomously AI-generated. The human provided:

- The overall architecture (which modules exist, how they connect).
- The mathematical strategy (Parseval Bridge over Vasyunin, Rank-1 Mellin over Hahn–Banach).
- Quality control (accepting, rejecting, or modifying AI-generated code).
- Editorial judgment (what to archive, what to keep, how to document).

This is analogous to a photographer using a camera: the camera captures the image, but the photographer makes the creative decisions (composition, timing, subject). Courts have consistently held that photographs are copyrightable despite the mechanical role of the camera.

Likely outcome: The Cathedral's code is copyrightable by the human author, as the human provided sufficient creative direction. The AI's contribution is analogous to a camera's or a compiler's: mechanical execution of human-directed creative choices.

2.3 Terms of Service Considerations

Both Google and Anthropic's terms of service (as of 2026) assign ownership of AI-generated output to the user. However:

- These terms could change.
- They may not be enforceable in all jurisdictions.
- They do not address the underlying copyright question (whether the output is copyrightable at all).

3 Question 2: Can AI Be a Co-Author?

3.1 Academic Authorship

The Cathedral's companion papers credit AI systems as "collaborative contributors." This raises questions under academic authorship standards:

- **ICMJE criteria**: Authorship requires (a) substantial contribution, (b) drafting or critical revision, (c) final approval, and (d) accountability. AI systems cannot satisfy (c) or (d).
- **ACM policy**: The ACM prohibits listing AI systems as authors but requires disclosure of AI assistance.
- **Nature/Science**: Major journals require that all authors be human and accountable.

The Cathedral's documentation credits AI as a contributor in the acknowledgments, not in the author line. This is consistent with current academic norms but may understate the AI's contribution (85% of code by volume, strategy proposals adopted by the human).

3.2 Inventorship

Under U.S. patent law (35 U.S.C. § 100), an inventor must be a "natural person." In *Tháler v. Vidal* (2022), the Federal Circuit held that AI cannot be named as an inventor on a patent.

The Cathedral's methods (Parseval Bridge, Rank-1 Mellin Miracle) are mathematical techniques, not patentable inventions. However, the engineering applications proposed in companion papers (Möbius Harmonic Analyzer, arithmetic compressed sensing, prime-spaced arrays) could potentially be patentable. If AI proposed these applications, the inventorship question becomes relevant.

Current law: Only the human can be named as inventor, even if AI substantially contributed to the inventive concept. This may change as legislatures address AI inventorship.

4 Question 3: Machine-Verified Proofs as Legal Evidence

4.1 The Evidentiary Question

Machine-verified proofs could serve as evidence in legal proceedings:

- **Patent disputes**: "Our algorithm is provably correct" (verified in Lean 4).
- **Safety certification**: "This bridge design meets load requirements" (verified by formal analysis).
- **Regulatory compliance**: "Our financial model satisfies Basel III constraints" (verified).
- **Product liability**: "The software was formally verified to be free of the defect that caused the accident."

4.2 Admissibility

Under the Federal Rules of Evidence (FRE 702), expert testimony must be based on "sufficient facts or data" and "reliable principles and methods." A machine-verified proof could qualify as:

- **Expert testimony**: The Lean compiler's output as an expert opinion on the correctness of a mathematical argument.
- **Documentary evidence**: The Lean source code as a document establishing a mathematical fact.
- **Computer-generated evidence**: Similar to DNA analysis or financial modeling output, subject to authentication (FRE 901) and the Daubert standard.

Key advantage: Unlike human expert testimony, machine- verified proofs are *deterministic* and *reproducible*. Any party can re-run the Lean compiler and obtain the same result. This satisfies the Daubert factors (testable, peer-reviewed, known error rate [zero], generally accepted in the relevant community).

Key challenge: The proof's correctness depends on its *axioms*. An opposing party could challenge the axioms rather than the proof. The Cathedral model—where every axiom is named, classified, and documented—facilitates this: the debate shifts from "is the proof correct?" (machine-checked) to "are the assumptions reasonable?" (human judgment).

5 Question 4: Open-Source Licensing

5.1 The Licensing Landscape

The Cathedral uses and depends on open-source components:

Component	License	Implication
Lean 4	Apache 2.0	Permissive, commercial use OK
Mathlib	Apache 2.0	Permissive, commercial use OK
Cathedral code	Not yet licensed	See below
Companion papers	Not yet licensed	See below

5.2 Licensing Considerations

The Cathedral's author must decide on a license. Options include:

1. **Apache 2.0 / MIT**: Consistent with Lean/Mathlib ecosystem. Allows commercial use and modification. Maximizes adoption. Does not prevent military or surveillance use.
2. **GPL**: Requires derivative works to be open-source. Limits commercial exploitation. Inconsistent with Mathlib's Apache 2.0 license (potential compatibility issue).
3. **Creative Commons (for papers)**: CC-BY for maximum dissemination. CC-BY-NC to prevent commercial exploitation of the documentation.
4. **Custom license with ethical restrictions**: Could prohibit use for specific purposes (e.g., surveillance, weapons). Difficult to enforce. Not OSI-approved. May limit adoption.
5. **No license (all rights reserved)**: Default under copyright law. Maximizes author control. Minimizes adoption and community contribution.

Recommendation: Apache 2.0 for code (ecosystem consistency), CC-BY-4.0 for papers (maximum dissemination with attribution). This is consistent with the project's stated values of transparency and open science, and with the companion papers' argument that open publication favors defenders over attackers.

6 Question 5: Liability

6.1 If the Proof Has an Error

The Cathedral claims zero compilation errors and two explicit axioms. But what if:

1. **Lean has a bug**: The Lean compiler itself could have a soundness bug that accepts an invalid proof. This is theoretically possible but highly unlikely (Lean's kernel is small and well-audited).

2. **An axiom is false**: One of the 42 axioms could be mathematically false. The proof would be "correct" conditional on a false assumption. The tiered axiom classification mitigates this: Tier 1 axioms are well-established results; Tier 4 axioms are structural claims that could potentially be wrong.

3. **A formalization error**: The Lean code could correctly implement a theorem that does not accurately represent the intended mathematical statement. This is the "specification bug" problem.

6.2 Liability Framework

If someone relies on a Cathedral-derived result (e.g., a verified stability certificate for a bridge) and that result is wrong:

- **Product liability**: Does the proof "product" create liability for the author? Under current law, probably not—mathematical proofs are information, not products. But if the proof is packaged as a commercial service ("certified correct by formal verification"), product liability could attach.

- **Professional liability**: If the author holds themselves out as providing engineering verification services, professional liability standards apply.

- **AI liability**: If the AI contributed to an error, who is liable—the AI provider (Google, Anthropic), the user, or the compiler? Current law does not clearly answer this.

Mitigation: The Cathedral's axiom transparency is a liability shield. Every assumption is documented. A user who relies on the proof accepts the axioms; if an axiom is false, the proof's warranty is limited to "correct conditional on stated axioms." This is analogous to an engineering report that states its assumptions explicitly.

7 Emerging Legal Questions

The Cathedral anticipates several legal questions that will become urgent as AI-assisted formal verification becomes common:

1. **Verification as due diligence**: Will formal verification become a legal standard of care? If a bridge *could* have been formally verified but wasn't, does the engineer bear liability for using a less rigorous method?

2. **Axiom disclosure**: If a verified claim depends on axioms, should there be a legal requirement to disclose them? (Analogous to financial disclosure requirements.)

3. **AI training on proofs**: If an AI system is trained on the Cathedral's code, does the Cathedral's license apply to the AI's output? (The "derivative work" question for AI training data.)

4. **Jurisdictional issues**: Mathematics is universal, but IP law is jurisdictional. A proof verified in the U.S. has no special legal status in the EU or China.

5. **National security**: Can a government classify a mathematical proof? The dual-use paper identifies verified instability certificates as potentially sensitive information. Current frameworks (CEII, ITAR) do not clearly cover mathematical proofs.

8 Conclusion

The Cathedral raises more legal questions than it answers. This is expected: the law follows technology, not the reverse. The key insight for legal professionals is that machine-verified proofs create a *new category* of intellectual output:

- More rigorous than expert opinion (machine-checked).
- More transparent than traditional proof (axioms named).
- More reproducible than simulation (deterministic compilation).
- Created by a novel collaboration (human + AI + compiler).

Existing legal frameworks—copyright, patent, evidence, licensing, liability—were designed for a world where intellectual work is produced by humans using passive tools. The Cathedral is produced by a human using active tools (AI) verified by an objective tool (compiler). The legal frameworks will need to adapt.

The recommendation to legal professionals: study the Cathedral's methodology now, before the first case arrives on your desk. The questions are coming. They are inevitable. The only question is whether the law is prepared.

The compiler does not practice law.
But it produces evidence that lawyers
will need to understand.

References

[1] S. Thaler v. R. Perlmutter, No. 1:22-cv-01564 (D.D.C. Aug. 18, 2023).

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[3] U.S. Copyright Office, *Copyright Registration Guidance: Works Containing Material Generated by Artificial Intelligence*, 88 Fed. Reg. 16190 (Mar. 16, 2023).

[4] Daubert v. Merrell Dow Pharmaceuticals, 509 U.S. 579 (1993).

[5] ACM, *Policy on Authorship*, updated 2023.